

The vanishing abnormal returns of momentum strategies and ‘front-running’ momentum strategies

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Abstract

We find large variations in returns from momentum strategies. Momentum strategies did not earn significant returns during the period of 1993-2004 which was due to their poor performance over the period from 2001-2004. We find that the previously documented large firm momentum effect is sensitive to the momentum strategy examined, and is in our sample driven by the abnormal returns of large Nasdaq stocks. We also evaluate momentum strategies that do not adhere to the end of month portfolio formation universally used in the academic literature. To this end we form portfolios one week prior to the end of month and call them ‘front-running’ momentum portfolios. Consistent with institutional momentum trading affecting end of month returns and volatility, we find that ‘front-running’ a momentum strategy generates similar, but less volatile returns than the month-end strategy. Pertinently, the returns of a ‘front-running’ strategy are consistently less volatile than that of an equivalent month-end strategy.

1. Introduction

We assess the profitability of momentum strategies in the twelve years since the seminal Jegadeesh and Titman (1993) study. The paper examines the profitability of momentum strategies in various subperiods and during specific market states to gain a better understanding of the relationship between the strategy and overall market conditions. To this end, we form Jegadeesh and Titman zero-cost momentum strategies using a comprehensive dataset that is adjusted for distributions and delistings. We then sort individual portfolio months by the prevailing market condition at the time. Various classifications of market conditions are used as a robustness check. The structure of

momentum returns is also investigated by examining the returns in size and exchange-based sub samples.

We then investigate whether the widespread use of momentum strategies following the discovery of the momentum effect has had any impact on stock returns. We note that if institutional momentum trading affects month-end stock prices, evaluating a momentum strategy prior to the month-end should generate returns in excess of that achieved by a regular month-end strategy. To this end we construct ‘front-running’ momentum strategies, strategies that evaluate momentum portfolios five business days before the month-end. Such strategies are otherwise identical to their month-end counterparts. To study the ‘front-running’ effect in greater detail, we investigate a number of sub samples and momentum strategies that prior literature has suggested will be most affected by institutional momentum trading. We then compare the return of these strategies with those of conventional month-end strategies.

In contrast to previous studies, we find that momentum returns are not significant for the entire sample period. We show that this is due to the period during and after the market crash experiencing abnormal momentum returns. In the sample period, momentum is related to stock market movements, but not to movements in the economy as a whole. Interestingly, we find that the large firm momentum effect is not robust to the specification of the ranking period and appears to be driven by the abnormal returns of large Nasdaq stocks within the sample period.

Consistent with institutional momentum trading affecting end of month returns, we find that ‘front-running’ a momentum strategy generates economically significant

returns in excess of a month-end strategy only for small firms. Further the returns of a ‘front-running’ strategy are consistently less volatile than that of an equivalent month-end strategy. The ‘front-running’ effect is most pronounced when using a three-month ranking period.

The seminal paper on momentum was Jegadeesh and Titman (1993), who show that trading strategies that buy stocks that had performed well in the past three to twelve months and sell stocks that had performed poorly in that period realize significant returns. They examine stock returns in the NYSE and AMEX over the 1965-1989 period, ranking these at the *end* of each month on their past performance over the last 1-4 quarters (the ‘*ranking period*’). The portfolio that equally weights the decile with the highest returns over the ranking period is then referred to as the ‘*winner*’ portfolio. Similarly, the portfolio of the decile with the lowest returns is referred to as the ‘*loser*’ portfolio. The relative strength or momentum strategies Jegadeesh and Titman examine involve subsequently buying the winner portfolio and selling the loser portfolio for a ‘*holding period*’ of 3, 6, 9 or 12 months. Thus, the momentum strategies are zero-cost, self-financing strategies. Refinements of the basic strategy include a gap or skip period between the ranking and holding periods to avoid contamination of the results by microstructure effects and the bid-ask bounce. [Jegadeesh (1990) and Lehmann (1990)] Jegadeesh and Titman report that the returns of all the momentum strategies they examine are positive and statistically significant, obtaining zero-cost profits of about one percent per month. The exception to this finding is a strategy that ranks stocks on past three month returns with a three month holding period and no skip period (a 3/0/3

strategy in standard momentum notation).¹ To examine the viability of momentum strategies Jegadeesh and Titman also examine the profitability of momentum strategies in size and beta subsamples. They find that the profitability of momentum strategies is not confined to any sub sample of stocks, with momentum strategies earning significant profits in small, medium and large firms, and across beta-based subsamples. Further, Jegadeesh and Titman find that the abnormal performance of momentum strategies is due to the long side of the portfolio, rather than the short side. This suggests that momentum profits are indeed obtainable.

Given the magnitude and significance of the momentum returns documented by Jegadeesh and Titman (1993), the issue of data mining was of initial concern. Indeed, the early literature and corresponding empirical evidence employed essentially the same database of US stocks listed on the NYSE and AMEX. As such, it could not be dismissed that the apparent momentum anomaly was not simply due to the process of “data snooping” as defined by Lo and MacKinlay (1990). In their original study, Jegadeesh and Titman addressed the data mining issue by examining the returns of a 6/0/6 strategy over five-year subperiods. They find that such a momentum strategy obtains positive returns in all but the 1975-1979 subperiod, with the negative returns in this subperiod primarily due to the January returns of small firms. When either January or small firms were excluded from the sample, momentum returns were positive in all subperiods. These results suggest that momentum is unlikely to be a result of data mining.

¹ The notation $J/S/K$ denotes the momentum strategy with a ranking period of J months and a holding period of K months. S denotes the length in months of the skip period (if any) between the ranking and holding periods.

In the late 1990's a number of studies examined momentum in markets outside of the US, providing further evidence of the profitability of momentum strategies.

Rouwenhorst (1998) examines the momentum return patterns of 12 European markets over the period of 1978 through 1995. Replicating the methodology of Jegadeesh and Titman (1993), he finds that an internationally diversified momentum portfolio generates similar returns to that documented by Jegadeesh and Titman, with a return of approximately one percent per month. The returns of all strategies examined were positive and significant. To address the concern that this result is driven by country momentum, Rouwenhorst examines the return of momentum strategies in each market, finding positive and significant returns in all countries examined in the sample with the exception of Sweden. This suggests that the momentum returns documented are not due to country momentum, and that momentum is indeed pervasive and not restricted to a few individual markets. As with Jegadeesh and Titman, Rouwenhorst finds that market risk is unable to explain momentum returns and that momentum profits are present in both large and small-capitalization firms. Due to the high correlation between the US market and the European markets examined, momentum could be explained by a factor common to these markets. To address this concern, one could examine momentum returns in markets that are not (highly) correlated with the US market. This issue is analyzed by Chui, Titman and Wei (2000), who examine momentum returns in eight different Asian countries: Hong Kong, Indonesia, Japan, Korea, Malaysia, Singapore, Taiwan and Thailand. As with Rouwenhorst (1998), Chui et al. examine the momentum effect when stocks from all 12 countries are included in one composite sample, finding a weak momentum effect. However, this result is primarily driven by the dominance of Japanese stocks, which

exhibit a very weak momentum effect. Excluding Japanese stocks, Chui et al. find that the momentum effect is strong and statistically significant, returning 1.45 % per month in the 1975-1997 subperiod². This suggests that momentum strategies are also highly profitable when implemented on Asian stock markets outside Japan.

In a more recent study, Griffin, Ji and Martin (2003, 2005) examine the profitability of momentum strategies in the 40 countries with more than 50 stocks listed on the Datastream International database. They find that momentum strategies are generally profitable across the countries examined, with momentum returns highly significant in all regions – i.e. Africa, the Americas (excluding the US), Europe and the US – except for Asia.³ Moreover, Griffin et al. find that momentum returns are distinctly lower in Asia than those around the world.⁴ As with Rouwenhorst (1998), Griffin et al. also find that momentum profits are lower in emerging markets, with the average momentum return of emerging markets an insignificant 0.27% a month. In comparison, the average momentum return of developed markets (excluding the US) was a significant 0.73% a month.

To examine whether the momentum anomaly is restricted to their original sample, Jegadeesh and Titman (2001) obtain nine additional years of data (relative to their 1993 study) and perform an out-of-sample test of their original findings. Using the Jegadeesh

² Chui et al. actually examine momentum returns over the 1975-2000 period. However, momentum returns were markedly different in the 1997-2000 subperiod after the Asian financial crisis., with an insignificant return of 0.530% per month.

³ Griffin et al. (2003) employ regional averages to allow for noisiness in individual country data. The regional average is simply the time series of the equally weighted average of all countries in the region.

⁴ Daniel, Hirshleifer, and Subrahmanyam (1998) suggest that the weak momentum effects observed in Asian markets is reflective of the low levels of biased self-attribution in these markets. See Section 2.4 for a discussion of the Daniel et al. model.

and Titman (1993) methodology, they find that momentum strategies are profitable over the post-sample period of 1990-1998, with a (significant) monthly return of 1.39%. This compares favorably to the sample period (1965-1989) return of 1.17%. Jegadeesh and Titman also find that the momentum effect has continued in the 1990s for both large and small stocks. As such, they conclude that momentum profits are not due to data snooping biases. Similarly, Griffin et al. (2003) find that momentum returns are profitable around the world in a sub sample of 1990-2000. These results suggest that momentum profits have not been extinguished since the original Jegadeesh and Titman (1993) paper and interestingly that market participants have not altered their investment strategies in a way that would eliminate momentum returns.

Given the extensive out-of-sample evidence that has documented momentum from both geographic and temporal perspectives, the presence and robustness of momentum returns is widely acknowledged. Indeed, Grinblatt and Moskowitz (2004) note that data biases cannot explain momentum returns. Nonetheless, it should be noted that the momentum effect only appears to be present in developed markets, with insignificant returns accruing to momentum strategies in emerging markets.

However, to date almost no research has been conducted on the pattern of momentum returns during the boom period of the late 1990's and the following market crash. This is despite the strong and distinct trends in the market over this period making it ideal for an examination of the pattern and structure of momentum returns. The exception to this is Boni and Womack (2004), who find insignificant momentum returns over the 1996-2002 period. However, Boni and Womack do not examine momentum returns in any great detail, with the focus of their study on analyst recommendations.

Furthermore, with a few notable exceptions (for example Grinblatt and Moskowitz, 2004 and Jegadeesh and Titman, 2001), prior momentum studies have generally examined the returns of strategies that select stocks from the NYSE and AMEX.

To increase the power of their tests and to allow simple t-statistics to be used, *overlapping* portfolios are utilized, whereby the investment strategy is followed every month, such that at any month t , a series of momentum portfolios are held from the previous $k-1$ months, where k is the holding period. The k individual momentum portfolios are then equally weighted.

While the presence of momentum is widely accepted, there is no consensus on the source and cause of momentum returns. Indeed, the presence of momentum is puzzling, as it suggests that prices are not even weak-form efficient. For the presence of momentum to be explained by rational risk-based models, risk would need to increase after positive returns, contrary to intuition which suggests that risk should decline. In this context, the explanation of momentum returns is subject to intense debate, with a voluminous literature documenting a variety of theoretical and empirical-based explanations. Essentially, this literature can be grouped into risk-based and behavioral-based explanations. This section will detail the potential sources of momentum profits. It will then examine the seminal Jegadeesh and Titman (1993) paper in context of these sources of momentum returns. The following sections will examine risk-based and behavioral explanations of momentum returns detailed in subsequent literature.⁵

⁵ Beyond the studies examined in this literature review, Berk, Green and Naik (1999), Johnson (2002), Lee and Swaminathan (2000), Jiang, Lee and Zhang (2004) and Grinblatt and Moskowitz (2004) detail alternative explanations for momentum returns.

Before examining the various explanations of momentum, it is useful to present a simple model of returns to illustrate the potential sources of momentum returns. Chordia and Shivakumar (2002) consider the following multi-factor linear process for stock returns:⁶

$$r_{it} = \mu_{it} + \sum_{k=1}^L \beta_{ik} f_{kt} + \sum_{m=1}^M \theta_{im} z_{mt} + e_{it} \quad (1)$$

where r_{it} is the return on security i , μ_{it} is the expected return on security i conditional on the information set at time t , f_{kt} is the return on the factor mimicking portfolio k (for instance the Fama-French factors), z_{mt} represents industry portfolio returns orthogonal to the returns on the factor-mimicking portfolios and e_{it} is the firm-specific component of return. β_{ik} is the factor loading of security i on factor k and θ_{im} is stock i 's sensitivity to the return on industry m .

Chordia and Shivakumar show that using the weighted relative strength strategy as defined by Lo and MacKinlay (1990)⁷ momentum returns equal:

$$\begin{aligned} WML = & \frac{1}{N} \sum_{t=1}^N (\mu_{it} - \bar{\mu}_t)(\mu_{it-1} - \bar{\mu}_{t-1}) + \sum_{k=1}^L \sigma_{\beta_k}^2 \text{Cov}(f_{kt}, f_{kt-1}) \\ & + \sum_{m=1}^M \sigma_{\theta_m}^2 \text{Cov}(z_{mt}, z_{mt-1}) + \frac{1}{N} \sum_{t=1}^N \text{Cov}(e_{it}, e_{it-1}) \end{aligned} \quad (2)$$

⁶ The discussion in this section is quite standard in the literature. Refer to Grinblatt and Moskowitz (1999), Jegadeesh and Titman (1993) and Jegadeesh and Titman (2002) for similar models.

⁷ The weighted relative strength strategy (WRSS) is a zero-cost strategy that weights stocks by their past returns less the past equal-weighted index return. Jegadeesh and Titman (1993) find that the WRSS is closely related to traditional equally-weighted momentum strategies, with a correlation coefficient between the returns of the WRSS and a equally-weighted 6/0/6 strategy of 0.95.

where WML is the return of a momentum strategy, N is the number of stocks in the market and $\sigma_{\beta_k}^2$ and $\sigma_{\theta_m}^2$ are the cross-sectional variances of the factor portfolio loadings and the industry loadings respectively.

Equation 2 suggests four potential sources of momentum profits. The first term, $(\mu_{it} - \bar{\mu}_t)(\mu_{it-1} - \bar{\mu}_{t-1})$ represents cross-sectional dispersion in conditional expected returns. If securities that experience relatively high (low) returns in the holding period are expected to have higher (lower) than average returns in the next period, then momentum strategies will generate positive returns. The second term $\sum_{k=1}^L \sigma_{\beta_k}^2 Cov(f_{kt}, f_{kt-1})$ represents serial correlation in the factors. If these factors are positively serially correlated, then momentum strategies will tend to pick stocks with high β 's when the conditional expectation of the factor portfolio return is high, generating positive momentum returns. The third term, $\sum_{m=1}^M \sigma_{\theta_m}^2 Cov(z_{mt}, z_{mt-1})$, is the serial correlation in the industry return components. The serial correlation in firm-specific components is represented by the last term, $Cov(e_{it}, e_{it-1})$. If the returns of momentum strategies are due to any of the first three terms, then they can be attributed to compensation for bearing systematic risk. In contrast, if momentum returns are due to the last term this would suggest that the market is inefficient.

The seminal paper by Jegadeesh and Titman (1993) examines the post ranking betas of relative strength portfolios, finding that the beta of the past losers is higher than that of the past winners. As the beta of the momentum portfolio is negative, Jegadeesh and Titman argue that momentum returns are not due to the cross-sectional dispersion of expected returns. However, this argument assumes that market risk is the only source of

cross-sectional dispersion of expected returns, an assumption which has subsequently been rejected. The following section will examine the effects of these other risk factors. Further analysis leads Jegadeesh and Titman to suggest that momentum returns are not due to serial covariance of factor returns or from a lead-lag relationship.

Rather, Jegadeesh and Titman suggest two possibilities: firstly, investors who buy past winners and sell past losers (i.e. positive feedback traders) may move prices temporarily away from their long-run values, causing prices to overreact. Alternatively, the market may underreact to information concerning short-term prospects (e.g. earnings forecasts) but overreact to long-term information. Jegadeesh and Titman do not investigate either hypothesis.

In a contemporaneous paper, Fama and French (1992) find that a firm's size and book-to-market (i.e. the ratio of the book value of equity to the market value of equity) can explain cross-sectional differences in average returns across stocks. Fama and French (1993) extend this result using a time-series regression, finding that a three-factor model consisting of beta, size and book-to-market factors is able to explain the cross-section of average returns. This led to one of the first attempts to explain momentum returns by Fama and French (1996). Fama and French tested whether a number of anomalies including momentum were consistent with their three-factor model. They find that while the three-factor model is able to explain the long-term reversal as documented by DeBondt and Thaler (1985) and the other anomalies examined⁸, it is not able to explain

⁸ Fama and French also examine returns to portfolios formed on the earnings/price ratio (E/P), cash flow/price ratio (C/P) and past sales growth. Prior research had shown that low E/P, low C/P and high sales growth firms experience lower future returns. Fama and French find that this is primarily driven by such firm's exposure to the book-to-market factor.

the momentum effect. As such, a momentum strategy's profitability cannot be explained by its unconditional exposure to the Fama and French three-factor model. Indeed, Carhart (1997) proposes a four-factor model that augments the Fama and French three-factor model with a momentum factor. They find that this model is a substantial improvement on the CAPM and the three-factor model.

Whether momentum returns can be explained by their dynamic exposure to the Fama and French three-factor model, or a two-factor model comprising the equally-weighted market index and a size factor is examined by Grundy and Martin (2001). Using NYSE and AMEX-listed stocks from 1926 to 1995, they form momentum portfolios and estimate the factor loadings of the momentum strategy each month. The risk-adjusted profit is then measured as the monthly return minus the product of the factor loadings and the respective factor return. After adjusting for the momentum strategy's Fama and French factor loadings, Grundy and Martin find that momentum returns increase from the raw return of 0.44% per month to 1.48% per month (with a t-statistic of 7.83). That is, the strategy earns an average risk-adjusted return of 1.48% per month. Even stronger results are obtained using the two-factor model. As such, Grundy and Martin show that momentum profits cannot be attributed to conditional exposure to the Fama and French three-factor model.

However, the cross-section of stock returns may be explained by factors other than the two-factor and three-factor models considered. To control for this, Grundy and Martin (2001) examine the mean-adjusted return, whereby the holding period return is adjusted by its time-series mean. They find that the mean-adjusted return is statistically indistinguishable from raw momentum returns. This suggests that the profitability of a

momentum strategy is statistically different from that in months outside the ranking period and that momentum profits cannot be attributed to expected returns.

Conrad and Kaul (1998) argue that the profitability of momentum returns could be explained by cross-sectional variation in unconditional expected returns. They argue that the higher returns of winners in the holding period represent their higher unconditional expected return, and similarly that losers are actually stocks with a low unconditional expected return. Conrad and Kaul perform a number of simulation and bootstrap experiments and find that momentum profits can be obtained with randomly generated data with no time series dependence. This result is inconsistent with the argument that momentum returns are generated by the time series pattern of returns since any patterns such as autocorrelation are eliminated with the bootstrap process.

As the Conrad and Kaul (1998) hypothesis argues that momentum returns are due to unconditional expected returns, an implication of their hypothesis is that momentum profits should persist in post-ranking periods. However, this is contrary to the event time analysis conducted by Jegadeesh and Titman (1993) in their seminal paper, which finds evidence of an ‘inverted U-shaped’ return structure. Specifically, momentum returns are found to be positive in months two to twelve after the formation period and negative on average in the second and third year after the portfolio was formed.

Jegadeesh and Titman (2001) examine this argument in greater detail, finding that over the 1965-1998 sample period, the momentum strategy experiences negative returns from month 13 to 60 (after positive returns in months 2-12)⁹. They find that the average

⁹ Negative returns are obtained in the first month, which reflects the short-term reversal.

return of -0.26% over this period is significantly less than zero, supporting the DeBondt and Thaler (1985) finding of long-term reversals. Further, Rouwenhorst (1998) and Griffin et al. (2003) document the existence of reversals throughout the world, finding that cumulative momentum profits generally peak after 6-10 months. Thereafter an intense reversal occurs, which Griffin et al. find to continue for at least 60 months after the portfolio formation date. This evidence of significant reversals in months 13-60 suggests that the source of momentum profits is not due to variation in unconditional expected returns. Indeed, as Jegadeesh and Titman note, this evidence is inconsistent with the Conrad and Kaul hypothesis and provides support for the behavioral models of Daniel, Hirshleifer, and Subrahmanyam (1998), Barberis, Shleifer, and Vishny (1998) and Hong and Stein (1999) which predict such a reversal. A more formal argument against the Conrad and Kaul (1998) hypothesis is presented by Jegadeesh and Titman (2002). Jegadeesh and Titman argue that the momentum profits generated by the Conrad and Kaul experiments arise due to a small sample bias. They show that when sampling with replacement, as employed by Conrad and Kaul (1998), the estimator of the cross-sectional variance is biased due to the effect of 'extreme' observations. In contrast, sampling without replacement yields unbiased estimators. Thus Jegadeesh and Titman replicate the Conrad and Kaul experiments using sampling without replacement, finding that the cross-sectional variance in returns is insignificant. This suggests that it is the time series variation in returns that generates momentum returns, rather than the cross-sectional variation in returns. Consequently, Jegadeesh and Titman conclude that the cross-sectional variance is able to explain at most a trivial proportion of momentum returns.

1.1. Industry momentum

Moskowitz and Grinblatt (1999) suggest that the profitability of individual-stock level momentum strategies is due to industry factors. Their approach involves using 2 digit SIC codes (i.e. the SIC Major Group) from the Center for Research in Security Prices (CRSP) to sort stocks into industries, forming market-capitalization (or value-weighted) industry portfolios. The industry momentum strategy examined takes a long position in the top three industries based on past six-month returns, and a short position in the bottom three industries. These portfolios are then held for six months. Moskowitz and Grinblatt find that such a strategy produces an average monthly return of 0.43%, identical to that of a stock-level momentum strategy.¹⁰ They find that after adjusting holding returns for size, book-to-market and industry effects, the returns of individual stock momentum strategies are eliminated. Conversely, the reverse is not true, with individual past stock return not accounting for industry momentum profits. Moskowitz and Grinblatt's findings thus suggest that industry momentum strategies dominate individual stock-level momentum strategies. They argue that momentum is not generated by serial covariation in firm-specific components, but rather industry components are the primary source of momentum profits. This suggests that the returns of momentum strategies are simply a compensation for industry risk as the winner and loser portfolios tend to be concentrated in a few industries.

However, the authors do not examine why industry momentum is able to explain stock-level momentum. Indeed, an examination of Moskowitz and Grinblatt's results

¹⁰ Moskowitz and Grinblatt examine the returns of a 6/0/6 strategy that takes a long (short) position in a value-weighted portfolio of the top (bottom) 30% of stocks based on past six-month returns.

suggests some inconsistencies with prior (momentum) literature. Pertinently, Moskowitz and Grinblatt document that the industry momentum effect is most profitable in the short-term, with a strategy that ranks industries based on their past one-month returns and holds the ‘winners’ and ‘losers’ for one month (i.e. a 1/0/1 strategy) earning 1.05% per month. In contrast, the momentum literature (e.g. Jegadeesh, 1990, Lehmann, 1990, Jegadeesh and Titman, 1993) has documented a short-term *reversal* effect, with Jegadeesh and Titman finding that the greatest momentum profits accrue in the medium term of three to twelve months. This suggests the need of further research in this area.

The relationship between stock-level momentum and industry momentum is also examined by Grundy and Martin (2001). Grundy and Martin replicate the Moskowitz and Grinblatt (1999) methodology, using the same sample period but applying cumulative excess returns (rather than total raw returns) to rank stocks in the ranking period.¹¹ They find significant returns using a strategy with 6 month formation and holding periods and no gap between the ranking and holding periods. As with Moskowitz and Grinblatt, the returns of a ‘random industry strategy’, whereby each initial firm within a winner/loser industry is replaced by a firm matched on its ranking return, are insignificant. This suggests that it is the industry component that is the source of the profitability of industry momentum strategies. However, when a gap period is present between the ranking and holding periods, the industry momentum strategy does not generate significant positive returns. Further, the industry momentum strategy does not outperform the random industry strategy. This suggests that microstructure effects may have a role in explaining

¹¹ The different ranking criteria will be discussed in section 5. In brief, Grundy and Martin (2001) argue that defining winners and losers on total returns is suboptimal, and prescribe the use of cumulative excess returns.

the profitability of industry momentum strategies. In addition, industry momentum strategies appear to be less profitable than stock-level momentum strategies, with a value-weighted industry momentum strategy generating returns of 0.89% in non-January months. In comparison, a comparable value-weighted stock-level momentum strategy earns 1.59% in non-January months. This suggests that whilst industries do exhibit momentum behavior, industry momentum is unable to explain stock-level momentum.

Lewellen (2002) finds that momentum is present in both individual stocks and industries as well as size and book-to-market sorted portfolios. The latter result suggests that momentum is more pervasive than previously thought, but is nevertheless not inconsistent with behavioral models. As with Grundy and Martin (2001), adjusting for size, book-to-market and industry effects does not eliminate momentum returns but rather accentuates them. Similarly, Lewellen finds that industry momentum cannot be explained by size and book-to-market factors. This suggests that industry and stock-level momentum are distinct.

The apparent inconsistencies in the industry momentum literature are examined by Scowcroft and Sefton (2005), who identify two separate groups of momentum studies. The first group such as Grundy and Martin (2001) use the US CRSP database and (generally) construct equally-weighted momentum portfolios. These papers find that momentum exists only at the stock-level¹². The second group Scowcroft and Sefton identify use value-weighted portfolios based on a large-cap universe. These studies, such

¹² Scowcroft and Sefton classify Grundy and Martin (2001) in this first group of studies using equally-weighted, broad stock universes. However, Grundy and Martin find the presence of both stock-level and industry level momentum effects. In addition, they examine both equal and value-weighted momentum strategies, finding significant returns to stock-level and industry-level momentum strategies using both weighting methods.

as O'Neal (2000) find that momentum is present at both the stock and industry level. As a result, Scowcroft and Sefton examine momentum using both a large-cap MSCI World universe as well as a broader Dow Jones Global Index sample over the period of 1992-2003¹³. Using the large-cap sample, they construct a momentum strategy long (short) the two best (worst) performing MSCI World sectors, with each industry portfolio weighted according to market capitalization. As with Moskowitz and Grinblatt (1999), they find that the returns of such an industry momentum strategy are comparable to that of a similarly constructed stock-level momentum strategy.

Scowcroft and Sefton (2005) then estimate a linear factor model for stock returns based on the Heston and Rouwenhorst (1995) model. This model allows the decomposition of stock-level momentum returns by decomposing the returns of individual stocks into components attributable to sector, country and company-specific factors. Using this model, Scowcroft and Sefton find that company-specific momentum comprises only 8% of stock-level momentum returns on average, with sector components accounting for 58% of returns and country components accounting for 34%. This suggests that at a large-cap level, industry factors are able to explain much of the profitability of stock-level momentum strategies. Conversely, using equally-weighted momentum portfolios and the broader Dow Jones universe, company-specific momentum accounts for 66% of stock-level momentum returns, with sector components accounting for 20% of returns and country components accounting for 15%. As using equal weights and a broader stock universe attaches greater weight to small-cap stocks, this result

¹³ It is worth noting that the Dow Jones Global Index is not as broad as the CRSP database. Indeed, despite its global nature, it tracks fewer stocks (approximately 5,500) than the US CRSP database (which covers approximately 7,000 stocks).

suggests that industry momentum is unable to explain stock-level momentum in smaller stocks. It is worth noting that the Heston and Rouwenhorst linear factor model has some shortcomings due to the simplifying assumptions Scowcroft and Sefton make. Nevertheless, to the extent that a firm's sector is not indicative of the stock itself (which is likely given that there are only 10 sectors in the MSCI World Index), these assumptions would bias against industry components explaining the returns of a momentum strategy.

Scowcroft and Sefton hypothesize that momentum is driven by the slow incorporation of news into stock prices. They argue that for large-cap stocks, unexpected earnings announcements are more likely to be interpreted as news on the prospects of the industry as a whole. As a result, the price of stocks within an industry will move in response to an unexpected earnings announcement from a large firm within that industry. Additional unexpected earnings announcements will trigger further reactions, leading to momentum in these large-cap stocks. In contrast, unexpected earnings announcements regarding small-cap stocks are unlikely to be interpreted as indicative of industry prospects, and as such industry momentum is unlikely to drive observed momentum effects. Whilst these hypotheses are unproven, Scowcroft and Sefton's results do suggest distinct sources of momentum profits in large and small-cap stocks. They also suggest that industry momentum is able to explain most of the profitability of stock-level momentum strategies when these are restricted to large-capitalization stocks. Nevertheless, momentum portfolios have been shown to be dominated by smaller stocks (see Jegadeesh and Titman, 2001, Grinblatt and Moskowitz, 2004), suggesting that at a domestic level where large-capitalization stocks comprise only a fraction of listed stocks

(and an even smaller proportion of the winner and loser portfolios) industry momentum is unable to explain momentum returns. This intuition is supported by Grundy and Martin's (2001) findings.

1.2. Momentum Strategies and Transaction Costs

The discussion in the previous sections highlights that there is no universally accepted, risk-based explanation of momentum returns. This suggests that momentum is evidence of market irrationality as abnormal profits can be obtained from such a simple trading strategy. However, Rubinstein (2001) points out that a market can still be efficient in the presence of such an anomaly. Rubinstein argues the need to define market rationality in a more flexible manner, with a market "*minimally rational*" if there are no abnormal profit opportunities for a rational investor. If an abnormal profit opportunity is found in such a market, Rubinstein argues that certain market imperfections exist to prevent investors from exploiting such an anomaly. As such, the apparent profit opportunity is only a paper profit and not actually obtainable. Similarly, Shleifer and Vishny (1997) argue that in some circumstances, arbitrage may be restricted, allowing prices to diverge from their fundamental values. As such, in the presence of significant "limits to arbitrage" market anomalies are not only likely to appear, but persist over time. In a related paper, Barberis and Thaler (2002) note that the presence of irrational traders may cause prices to deviate from their fundamental value. In such circumstances, the presence of fundamental risk, noise trader risk and implementation costs results in costly and risky arbitrage strategies, allowing the mispricing to persist. As such, a pertinent question is whether there are significant limits to arbitrage that prevent investors from trading and eliminating momentum profits. These limits to arbitrage would then be

sufficient to explain the persistence of momentum. The issue of interest is thus whether momentum strategies are actually profitable after transaction and other costs.

This issue is first addressed by Jegadeesh and Titman (1993), who argue that momentum returns exceed the transaction costs necessary to implement the strategy. They find that after considering a 0.5% one-way transaction cost, a 6/0/6 momentum strategy earns a significant risk-adjusted return of 9.29% per year. Similarly, Hong, Lim and Stein (2000) argue that transaction costs are insufficient to eliminate momentum returns.

Prior to Jegadeesh and Titman's (1993) initial finding on the existence of momentum returns, Grinblatt and Titman (1989) show that mutual funds tend to purchase stocks that had increased in price over the past quarter. Grinblatt et al. (1995) extend this analysis by obtain quarterly portfolio holdings for 274 mutual funds and examining the extent of momentum trading and herding behavior. They construct a measure of momentum trading based on stock returns contemporaneous to portfolio revisions, finding that on average, the stocks held by a fund at the end of a quarter had returns a highly significant 0.74% greater than the stocks held at the start of the quarter. Further, the stocks bought had returns approximately 2% higher than those sold. This indicates that mutual funds engage in momentum trading, with further analysis revealing that 77% of mutual funds buy past quarter winners and/or sell past quarter losers. For the lagged (i.e. previous) quarter, Grinblatt et al. find that only 30% of funds engage in momentum trading. This suggests that mutual funds attach greater importance to the immediate past period return and not returns from prior quarters. Grinblatt et al. find that momentum

investing is primarily due to mutual fund buying past winners, rather than selling past losers. Further, the momentum trading is dominated by large capitalization winners.

A concern with the Grinblatt et al. findings is that the study is based on a relatively small sample of just one type of institutional investor. In addition, the age of their dataset (it covers the period of 1975-1984) raises concern about the applicability of their results. More recently, Wermers (1999) finds evidence of institutional momentum trading. However, this study is also restricted to mutual funds. Badrinath and Wahal (2002) address both of these concerns by obtaining quarterly holdings data from *all* investment managers (including banks, advisers, pension funds, insurance companies and investment companies) over the period of 1987-1995. They find that institutions tend to buy stocks that had enjoyed consistent over-performance over the past year and sell stocks with above average performance. This suggests that institutions simultaneously act as momentum buyers and contrarian sellers. Badrinath and Wahal also find evidence that the primary mechanism employed by institutions to adjust their portfolio is entry and exit, rather than through adjusting the level of their holdings.

In addition to engaging in momentum trading, Grinblatt (1995) find that the average fund engages in herding behavior. Specifically, mutual funds bought (sold) stocks that were subject to excessive buying (selling) by all funds on average. As funds engage in momentum trading, Grinblatt argue that herding in past winners occurs. Wermers (1999) also finds evidence of herding, particularly in small stocks. On the other hand, herding is weak for (actively traded) large stocks. Significantly, he finds that herding is most pronounced for stocks with extreme past returns, suggesting that funds are likely to not only buy prior winners and sell prior losers, but that these stocks are

subject to substantial herding effects. Wermers finds that this effect is related to positive feedback trading strategies.

Given that the literature indicates that institutions engage in momentum trading and that herding is present for momentum stocks, an interesting question is whether institutional momentum trading influences the return of momentum stocks. To this end, DeLong, Shleifer, Summers and Waldmann (1990) propose a positive feedback model. In this model irrational positive feedback traders behave in a trend-chasing fashion, resulting in prices overreacting to news. As such this model argues that momentum trading causes momentum effects. Similarly, the aforementioned Hong and Stein (1999) information diffusion model argues that trend-chasing momentum traders cause prices to overreact if information diffuses gradually across the population. Empirically, Wermers (1999) finds that there is a strong contemporaneous relation between changes in institutional trading ownership and stock returns. This relationship is found to be most pronounced for small stocks. Goetzmann and Massa (2003) also find evidence that mutual fund herding behavior impacts on stock returns. Indeed, as noted by Kumar (2005), “a growing number of studies suggest that coordinated demand shifts...can affect asset returns”. Despite this, to date no research has specifically examined whether institutional trading affects the returns of momentum stocks, nor constructed strategies to exploit such effects (if they exist).

2. Methodology

2.1 Data

To increase the power of their tests and to allow simple t-statistics to be used, *overlapping* portfolios are utilized, whereby the investment strategy is followed every month, such that at any month t , a series of momentum portfolios are held from the previous $k-1$ months, where k is the holding period. The k individual momentum portfolios are then equally weighted.

Our dataset consists of daily stock return data obtained from the Center for Research in Security Prices (CRSP). The sample is then constructed from all stocks traded on the New York Stock Exchange (NYSE), American Stock Exchange (AMEX) and Nasdaq stock exchange. To ensure that our results are not distorted by securities with abnormal return behavior, we employ a number of data restrictions. We include ordinary shares and exclude ADRs, REITs, closed-end funds, primes and scores and foreign incorporated companies. Specifically, we include stocks with CRSP share codes of 10 and 11. The use of similar restrictions is common throughout the literature, with Hong, Lim and Stein (2000) and Lee and Swaminathan (2000) employing identical data screens.

We obtain monthly holding period returns for each security by compounding the daily returns recorded by CRSP. The monthly returns thus obtained take into account all distributions received by stockholders and assume that these are reinvested when they are received. Further, our return data is adjusted for split events to ensure that the stock price data is directly comparable at different times during the history of a

security. These split events include stock splits, stock dividends and other distributions with price factors such as spin-offs, stock distributions and rights. In order to classify and sort stocks, we obtain descriptive variables from CRSP. Specifically, we obtain exchange data – defined as the exchange on which a security was last listed and the number of shares outstanding. The market capitalization of a stock is then defined as the market value of one stock multiplied by the number of shares outstanding. Market return data was also obtained from CRSP and is defined as the periodic return on the Standard & Poor's 500 Composite Index (S&P500). Finally, we obtain the monthly growth in real Gross Domestic Product from the U.S Bureau of Economic Analysis. These figures are seasonally adjusted at annual rates.

In order to examine the effects of momentum trading on stock prices, we examine the period subsequent to Jegadeesh and Titman's (1993) seminal momentum paper. That is, we consider momentum strategies from 1993 onwards. As any momentum strategy involves a ranking period, and due to the use of overlapping portfolios, we need to obtain an additional twenty-five months of stock data in order to construct our momentum portfolios.¹⁴ As such, our sample covers the period of December 1990 through to December 2004. Due to our analysis of momentum returns post-1993, the actual sample period used differs according to the momentum strategy (i.e. the length of the ranking and holding periods) employed.

¹⁴ The actual momentum strategies we examine will be discussed in the next section. For now, it is worth noting that a momentum strategy that ranks stocks on their past 12 month return, and holds these for 12 months with a 1 month gap period in between the ranking and holding periods (i.e. a 12/1/12 strategy) requires lagged returns data over 24 months. Using a front-running strategy will require data from an extra month.

The use of distribution-adjusted data appears to be relatively uncommon amongst previous literature, which tends to use unadjusted data. However, the use of raw stock return data fails to account for the distinctive return distribution of stocks affected by distribution events. Stocks are likely to experience ‘extreme’ returns in periods that include such events, and as a result are more likely to be included in the momentum portfolio. More problematic is the effect of such extreme returns for stocks that are within the momentum portfolio, with stock splits in particular having the potential to contaminate the return of momentum portfolios. Indeed, Grundy and Martin (2001) note that this is a potential problem, with one particular reverse stock split reducing momentum profits by -40.625% in that investment month.

It has been noted that the CRSP database does not report monthly prices or returns for stocks that were delisted in that month. In addition, Shumway (1997) documents a delisting bias in the CRSP database.¹⁵ However, in spite of these problems, delisting biases are rarely examined or addressed in the momentum literature. An exception to this convention is the work of Grundy and Martin (2001), who address the omission problem by eliminating stocks that are delisted after the ranking period but before the end of the holding period.¹⁶ This restriction implicitly assumes that investors possess perfect foresight of delistings, and will not consider investing in such stocks. Grundy and Martin note that when a winner is delisted, it is generally the result of a merger or takeover. In contrast, stocks within the loser portfolios are typically delisted due to liquidation or other negative performance-related reasons. Given the extensive empirical evidence that

¹⁵ The delisting bias inherent in the CRSP database will be discussed shortly.

¹⁶ Bar Grundy and Martin (2001) and Rouwenhorst (1998), the momentum literature does not appear to address or examine possible delisting biases. Indeed, the author could find no mention of delisting biases in the remainder of the literature examined for this study.

the share price of target firms increases substantially from a predefined period before the bid announcement date to the completion of the takeover (see Jensen and Ruback, 1983, Franks and Harris, 1989), this suggests that Grundy and Martin's assumption induces a bias against finding momentum returns.

Our usage of the CRSP daily data files removes an element of the bias inherent in the CRSP monthly database. Within the daily data files, daily return data is recorded for securities until they are delisted, with the last recorded daily trading value present in the database. However, when a stock is delisted, any return subsequent to the delisting is still ignored. To address this problem and the consequent bias it generates, we obtain delisting data from CRSP. For each delisted stock, we obtain the "delisting return", which is the return of a security after it is delisted from an exchange. This return is calculated by comparing the security's "amount after delisting" - the value of a security after it delists from an exchange, with its price on the last day of trading. The amount after delisting can itself be the off-exchange price, an off-exchange price quote, or the sum of a series of distribution payments. In our dataset, there are 8,545 stocks that are delisted during the sample period, representing 52.44% of the stocks in the dataset. The large number of delistings in the sample suggests that the CRSP database does not suffer from the survivorship biases that other databases such as the Datastream database experience.¹⁷

For each delisted stock, we examine the delisting return and add this to the sample if the date at which the amount after delisting is recorded is within the month the stock is

¹⁷ This is not to say that a survivorship bias free sample cannot be constructed using the Datastream database. Indeed, Datastream provides dead stock files that can be applied to recreate a complete sample. However, unadjusted and in its raw state, the relative paucity of delistings in the Datastream database indicates that it suffers from survivorship bias.

delisted. For example, if a stock is delisted in January 2000, CRSP will not record an end-of-month price for that stock in January 2000.¹⁸ If an amount after delisting is recorded in January 2000, we include the corresponding delisting return in our return calculations. Where the amount after delisting is recorded after the month the stock is delisted (for our example, in February 2000 or later), we do not include the delisting return in our return calculations, but rather set the monthly return to zero. There are 552 such firms. For these firms, the next available price may not be realized for months if not years and is thus generally highly uncertain.

As aforementioned, Shumway (1997) notes that the CRSP data files contain a delisting bias, finding that the CRSP files are missing thousands of delisting returns. The omission of returns from the CRSP database introduces a bias into studies that use the database, as the return of such stocks would be incorrectly calculated using the last trading price. Shumway finds that the delisting bias is primarily concentrated in firms that delist for bankruptcy, insufficient capital and other performance-related reasons, which are generally unanticipated. For such firms, the delisting return is likely to be close to -1. Although our system does not address the issue of delisting bias in its entirety, it does generate a more accurate and realistic dataset and more accurate results. Our system acknowledges the delisting bias inherent in the CRSP database and the uncertainty of delisting returns by assuming that no return is obtained within the month of delisting or after a delisting, if an amount after delisting is not available within the delisting month. As previous research has identified that this delisting bias is concentrated amongst stocks delisted due to negative performance reasons and that these stocks lie primarily within

¹⁸ Note that this example is for illustrative purposes only. As noted previously, we do not use the monthly CRSP files, but rather the daily data files.

the loser portfolio, we induce a bias against finding momentum returns. Nevertheless, the number of affected stock months is limited, with only 552 stocks not recording a delisting return within the delisting month. If we assume that all of these stocks are liquidated and subsequently worthless, the delisting bias as documented by Shumway (1997) will affect only 0.04% of our sample. Further, the proportion of firms that are subsequently worthless is unlikely to approach 100% of the stocks without immediate delisting returns, and as such, any delisting bias is at best minimal.

2.2 The Construction of Momentum Portfolios

The month-end momentum or relative strength portfolios are constructed using the methodology of Jegadeesh and Titman (1993). At the start of each month, the *ranking* or *formation period* return of each stock is calculated as the compounded total return over the past J months. The stocks are hence ranked on:

$$R_{it} = (1 + r_{i,t-J})(1 + r_{i,t-(J-1)}) \dots (1 + r_{i,t-1}) \quad (3)$$

where r_t is the month t return on stock i and J is the length of the ranking period in months. From the universe of stocks, those that have a return history of at least J months and were actively traded at both the start and end of the formation period are ranked on their formation period return. The restriction of the sample to actively traded stocks does not introduce a bias since all information necessary is known prior to the investment period.

The stocks are then split into deciles, with the best performing decile referred to as the ‘*winner*’ portfolio (P1). Correspondingly, the worst performing decile is referred to

as ‘*loser*’ portfolio (P10). Stocks within each portfolio are equally-weighted, and the relative strength or momentum strategies we examine involve subsequently buying the winner portfolio and selling the loser portfolio for a *holding period* of K months. As such, the momentum strategies are zero-cost, self-financing strategies.

Previous research (see Jegadeesh, 1990 and Lehmann, 1990) has documented the existence of reversals in the very short-term. Although this finding is robust, Lo and MacKinlay (1990) show that the existence of short-term reversals is a product of bid-ask bounce. If the spread is sufficiently large, Lo and MacKinlay argue that a short-term reversal will result. Evidence of the link between bid-ask spread and short-term reversals is provided by Jegadeesh (1990). The existence of short-term reversals may therefore contaminate intermediate-term momentum returns as the direction of the reversal is opposite to that of the momentum effect. To address this, strategies with a one month gap between the ranking and holding periods are also considered. Both sets of results are reported in the following section. With regards to the specific momentum strategies examined, past research, including that of Jegadeesh and Titman (1993) and Rouwenhorst (1998) show the results of different momentum strategies are quite similar, with average returns of approximately 1 percent per month when a ranking period of 6, 9 or 12 months is used regardless of the length of the holding period. Consequently, we only consider strategies that use either a three-month or six-month ranking period. In all, this yields 16 different strategies.

Consistent with the vast majority of the momentum literature, these momentum strategies all utilize overlapping portfolios, whereby the investment strategy is followed every month, such that at any month t , a series of momentum portfolios are held from the

previous $k-1$ months. In essence, this entails that for every month t , the strategy buys the winner portfolio and sells the loser portfolio, as well as closing out the position initiated in month $t-K$. For example, during December 2000, the 3/0/3 strategy consists of three different portfolio: the portfolio initiated in December employing a ranking period of September to November 2000; that initiated in November 2000 based on ranked returns from August to October 2000; and that initiated in October 2000 on the basis of returns over the period of July to September 2000. The k individual momentum portfolios are equally-weighted within the momentum strategy. Overlapping portfolios are used to increase the power of our tests and allows simple t-statistics to be used.¹⁹ Given the volatility of momentum returns, the use of overlapping portfolios also reduces the volatility of momentum returns.

With the passage of time, the value of the individual momentum portfolios will change, with those portfolios that had performed well during the holding period holding greater value. To address this issue, Jegadeesh and Titman (1993) use a strategy whereby the momentum portfolios are rebalanced monthly so that they continue to be equally weighted. This rebalancing would involve selling a proportion of those portfolios with above average performance (relative to the other momentum portfolios) during the portion of the holding period that has elapsed, and with the proceeds buying the poorer performers. If stock prices exhibit momentum, the sold portfolios will be expected to experience stronger performance in the future than those bought. Consistent with this argument, Jegadeesh and Titman find greater returns for a buy-and-hold strategy relative to the strategy where the series of momentum portfolios are rebalanced monthly.

¹⁹ Moskowitz and Grinblatt (1999) argue that the use of overlapping portfolios also reduces the effects of the bid-ask bounce, thereby providing cleaner results.

Nevertheless, the difference between the strategies is negligible, a result not unexpected given the minor difference in methodology. Accordingly, much of the previous literature examines the returns of momentum strategies that rebalance the individual momentum portfolios monthly. We will also follow the strategy of monthly rebalancing. As such, the return of the momentum strategy in any month t is the average return of the k individual momentum portfolios held in that month.

The holding period return of each individual momentum portfolio is calculated as the average of the stocks within the portfolio (as the portfolios are equally-weighted). However, as was noted in the previous section a number of stocks within the sample are delisted during the sample period. Where such stocks are included in momentum portfolios, no additional returns are attributed to that stock for the balance of its holding period. This is the most conservative method of handling delistings without introducing a selection bias. Other methods are to invest the liquidating proceeds in a market index or to invest the liquidating proceeds in the remaining stocks within the winner/loser portfolio. Employing the most conservative technique results in a small bias against finding momentum returns.

2.3 Momentum and Market States

To examine the pattern of momentum returns, we note that the behavioral models of Daniel et al. (1998) and Hong and Stein (1999) suggest that higher momentum profits

are gained in periods following market gains.²⁰ Cooper et al. (2004) examine this hypothesis, finding that momentum returns depend on the state of the market, with a positive and significant return of 0.93% following an ‘up-market’ and an insignificant return of -0.37% following a ‘down-market’. An up-market is defined by Cooper et al. as one where lagged three-year market returns are positive and a down-market as one where lagged three-year market returns are negative. The use of three-year lagged returns is arbitrary selected by Cooper et al., who find that the specification of market states is robust to using one or two year lagged returns.

To examine the relationship between momentum and overall market conditions, we replicate Cooper et al.’s methodology and classify each month in the sample period using its market state. As our sample period consists of 12 years, the use of lagged 3-year returns will result in the state of the market (henceforth the market state) to be defined in relation to one-quarter of the sample period. In addition, the use of a shorter timeframe may be more appropriate to capture the market state in the volatile market conditions of the sample. As such, we define lagged returns using compounded S&P500 returns over the previous year. The momentum portfolios are then constructed for up-market and down-market sub samples.

²⁰ The model of Daniel et al. (1998) suggests that the momentum effect is greater when investor overconfidence is high. To the extent that investor overconfidence is greater in periods following market gains, this suggests greater momentum returns in periods following stock market increases. Hong and Stein (1999) find that decreasing risk aversion leads to greater delayed overreaction and therefore greater momentum profits. If risk aversion decreases as wealth increases, the model of Hong and Stein also suggests momentum profits are higher in periods following market increases.

2.4 ‘front-running’ Momentum Portfolios

To date, academic research on the momentum effect has focused on portfolios formed at the end of the month. The ‘front-running’ relative strength portfolios are constructed in a similar fashion to that of the regular, month-end momentum portfolios. The fundamental difference between these portfolios and the month-end portfolios is the portfolio evaluation date. Conventional month-end momentum strategies rank stocks on their prior return at the end of each month, such that portfolio positions can be entered at the start of the following month. In contrast, the ‘front-running strategy’ involves evaluating ranking returns a period of five business days before the end of the month. The portfolio positions are then entered into the next business day, five days before the start of the following month. The stocks are hence ranked on the following criteria:

$$R_{i\tau} = (1 + r_{\tau-J})(1 + r_{\tau-(J-1)}) \dots (1 + r_{\tau-1}) \quad (4)$$

where r_{τ} is the compounded daily return on stock i for the month ending five business days before the end of the month and J is the length of the ranking period in months. For example, during April 2004, a normal, month-end momentum portfolio using a 6/0/6 strategy ranks the universe of common stocks with at least six months of return data over the period of 31st October 2003 to the 30th April 2004, holding the resultant momentum portfolio from 1st May 2004 to the 29th October 2004. The corresponding ‘front-running’ momentum strategy would rank the universe of stocks that satisfy the above criteria on their performance over the period of 24th October 2003 to the 23rd April 2004, holding the resultant momentum portfolio from 26th April 2004 to the 22nd October 2004.

2.5 Size Subsamples

As prior research has found that the momentum effect is present in both small and large-cap subsamples, of interest is whether this has continued in the sample period. As noted by Jegadeesh and Titman (2001), the robustness of momentum to size subsamples would provide strong evidence that momentum strategies are actionable. If however, momentum returns are not present in a specific sub sample, this may indicate that institutional trading has eliminated momentum returns for those stocks. As liquidity risk and transaction costs may impede the usage of momentum strategies for smaller stocks, the presence of momentum returns in large stocks would provide stronger evidence of the momentum effect.

Size subsamples are also used to investigate whether the ‘front-running’ effect is greater for any sub sample of stocks as per our hypothesis. We sort stocks into three groups, small capitalization stocks, mid-capitalization stocks and large capitalization stocks with reference to the NYSE market capitalization deciles. Specifically, at the portfolio evaluation date, firms with a market capitalization that would place it in the two lowest market capitalization deciles of listed NYSE stocks are classified as small stocks. Firms with a market capitalization above the median NYSE-listed stock are classified as large stocks, with the remainder classified as medium stocks.²¹

²¹ Alternatively, we could sort the stocks such that they contain a set proportion of the sample. However, as our sample includes Nasdaq and AMEX stocks, such a classification would result in the middle subsample consisting almost entirely of small capitalisation stocks as Nasdaq and AMEX stocks are generally small. Approximately 60% of the stock months in our sample are for stocks with market capitalisations that would place them in the two lowest NYSE deciles. The use of market capitalisation deciles thus results in more distinct subsamples.

2.6 Overlap of Month-End and ‘front-running’ Momentum Portfolios

To examine the effect of institutional momentum trading more explicitly, we obtain the ‘front-running’ and month-end momentum returns for those stocks common to both portfolios. As the sample of stocks included in the resulting ‘front-running’ and month-end strategies are identical, any difference in returns can be attributed solely to the return pattern of momentum stocks. As such, we are able to discern whether there is a distinct pattern of returns at the month-end for momentum stocks. If institutional momentum trading systematically influences month-end stock returns, we would expect that the ‘front-running’ strategies would generate larger returns than equivalent month-end strategies.

To investigate the stocks common to month-end and ‘front-running’ momentum strategies, we match each monthly ‘front-running’ winner or loser portfolio to the corresponding month-end winner or loser portfolio. Those stocks that are bought/sold by both strategies in a particular month are the ‘overlapped’ stocks. Consequently, the ‘overlapped stocks’ ‘front-running’ and month-end momentum portfolios consist of these overlapped stocks.²² The overlapped stocks month-end and ‘front-running’ momentum strategies are then constructed as before.

3. Results

Firstly, we find that momentum returns are not significant in the whole sample period. This is due to the period during and after the market crash experiencing abnormal

²² Note that the winner and loser portfolios hence constructed are not necessarily equal in size.

momentum returns. In the sample period, momentum is related to stock market movements, but not to movements in the economy as a whole. Interestingly, we find that the large firm momentum effect is quite sensitive to the specification of the sample and strategy. We also find that ‘front-running’ a momentum strategy generates economically if not statistically significant returns in excess of a month-end strategy for small firms. Further the returns of a ‘front-running’ strategy are consistently less volatile than that of an equivalent month-end strategy. The ‘front-running’ effect is most pronounced when using a three-month ranking period.

3.1 Momentum In The Entire Period

Table 1 presents the average monthly returns from several price momentum strategies over the period of 1993 to 2004. We report results for the bottom decile portfolio (R1), the top decile portfolio (R10) and the returns of a winner minus loser portfolio. Panel A in table 1 shows that across the entire sample period, past winners outperform past losers by about one percent a month although there is substantial variation in the profitability of individual momentum strategies. To illustrate, a strategy that ranks stocks on prior three-month returns and holds the resulting portfolios for three months (i.e. a 3/0/3 strategy) earns 0.28% per month, while a 6/1/3 strategy earns 1.01% per month.

Nevertheless, the returns generated by momentum strategies in our study are comparable with those of previous literature. For example, the oft-studied strategy that ranks stocks on the previous six month returns and subsequently holds the momentum portfolio for the following six months (i.e. a 6/0/6 strategy) returns an average of 1.01% per month. In comparison, Jegadeesh and Titman (1993) report a corresponding return of

0.95% per month over the 1965 to 1989 period, whilst Jegadeesh and Titman (2001) report a return of 1.39% per month over the 1990 to 1998 period. These results thus suggest that price momentum is present in our sample.

Insert Table 1 here

In contrast to previous studies, we find that the returns generated by momentum strategies are not statistically different to zero, with a t -statistic of 1.35 for the 6/0/6 strategy. In comparison, Jegadeesh and Titman (1993) and Jegadeesh and Titman (2001) report t -statistics of 3.07 and 4.71 for the 6/0/6 strategy. The statistical insignificance of our results is despite the obvious economic significance of the returns reported, with zero-cost returns of up 1.01% a month. A closer examination of our results suggests that this may be attributed to the highly volatile momentum returns in our sample, with the standard deviation of momentum returns using a 6/0/6 strategy 9.01%. Other strategies have similar (if not higher) standard deviations. Figure 1 plots the monthly time-series of returns for the 6/0/6 strategy. From the figure, it is readily apparent that the momentum returns in the sample period are highly volatile explaining the statistical insignificance.

Insert Figure 1 here

The portfolios in Table 1 are formed at the end of the ranking period. Because bid-ask bounce and other market microstructure effects are likely to attenuate the momentum effect, we examine the returns of corresponding strategies that contain a

‘skip’ period of one month between the end of the ranking and the start of the holding period. Previous research (for example see Jegadeesh and Titman, 1993 and Rouwenhorst, 1998) has shown that such strategies are superior to those that hold the momentum portfolios immediately. Specifically, strategies that wait a period of one month before holding the momentum portfolio have been shown to generate higher returns and are less volatile than strategies that do not skip a month. As such, the significance of these strategies would indicate that short-term microstructure effects and return reversals are contaminating the intermediate-term price continuation effect. The results of strategies that wait a period of one month between the ranking and holding period are reported in Table 6.2.

Consistent with prior literature, we find that momentum returns are greater when a skip period is included in panel B. Nevertheless, the returns of all momentum strategies considered are insignificant at the 5 percent level. We find that a 6/1/6 strategy generates an average return of 1.17% per month (t -statistic = 1.669) which is significant at the 10 percent level. The standard deviation of returns for this strategy is 8.43%. The results therefore suggest that although short-term effects attenuate momentum effects in our sample, they cannot explain the insignificance of momentum returns in the sample period.

3.2 Subperiod Analyses

The results of the previous section indicate that while momentum strategies earn large returns during the sample period, these returns are not statistically distinguishable from zero for our sample period. This finding is in stark contrast to the bulk of the existing literature, with few if any prior studies documenting insignificant momentum

profits in the US market.²³ However, this result is not entirely unexpected, with the sample period characterised by a bull market that resulted in a period of abnormally large sustained stock market increases, followed by the subsequent market crash after September 2000. The strong stock market trends in the sample period are most pronounced amongst high-technology firms, which experienced unprecedented growth in the late 1990's and early 2000, a period commonly referred to as the 'tech-boom'. The subsequent crash in technology stocks beginning in March 2000 resulted in the Nasdaq declining almost 80% over the following eighteen months.

3.2.1 Portfolio returns over 4-year subperiods

To investigate the effects of the tech crash on momentum returns, we examine momentum returns over four-year subperiods. Table 2 documents the returns of the 6/1/6 strategy in each of three four-year subperiods in the 1993 to 2004 sample period. Our results indicate that momentum returns are positive and significant in the 1993 to 1996 and 1997 to 2000 periods. However, during the 2001 to 2004 period, momentum returns are negligible, with a winner minus loser portfolio returning 0.08% over this period. This indicates that the insignificant momentum result for the entire sample period is a result of the non-existent return subsequent to the tech crash.

Insert Table 2 here

²³ We are aware of only two studies that have documented insignificant momentum returns in the US. In the first, Chen et al. (2002) find that an equally-weighted 12/0/12 momentum strategy does not generate excess returns to the market over the 1963 to 2000 subperiod. In a more recent study, Boni and Womack (2004) find that an equally-weighted 6/1/6 strategy does not earn significant returns in the 1996 to 2002 period. But their paper focuses on the information value of analyst recommendations.

3.2.2 Momentum and market states

To further investigate why momentum returns are not present in the 2001 to 2004 subperiod and to examine the pattern of momentum returns, we classify each month in the sample period as either an up-market or down-market. Panel A of Table 3 reports the average return of a 6/1/6 strategy in each market state.²⁴ Using this specification, the average momentum return in the 107 months defined to be an up-market is 1.23% per month, which is significant at the 5 percent level. Conversely, the average momentum return in the 37 months defined to be a down-market is an insignificant 1.01% per month. This suggests that momentum returns are related to the performance of the market as a whole. As expected, the down-market periods are highly concentrated in the post-tech crash period, with the months between November 2000 and June 2003 accounting for 87% of the down-market observations.

Insert Table 3 here

Following Griffin et al. (2003), we further examine the relationship between momentum and market states by dividing the time series of lagged market returns into quartiles. As reported in Table 3 Panel A, the average momentum returns are 1.08%, 0.56%, 1.86% and 1.19% for the lowest to highest quartiles of lagged market performance during the sample period. Of these, only the third quartile return is significant, suggesting that momentum strategies are most profitable when the market is

²⁴ Similar results are obtained using 2-year lagged returns.

steadily rising. More importantly, the insignificance of momentum returns in the first two quartiles is consistent with lower momentum returns accruing in weaker market states. Nevertheless, if momentum is caused by a behavioral biases of investors, classifying market returns into quartiles is problematic as it assumes that the response of investors to the market state is comparable across the whole quartile of lagged returns. Rather, the documented loss aversion amongst investors suggests that the impact of negative returns on investor overconfidence and risk aversion will be far greater when lagged returns are negative relative to when lagged returns are positive. As such, the grouping of periods of negative lagged returns with those of small positive returns is inconsistent with the behavioural models that underlie the hypothesised relationship between the market state and momentum returns.

Nevertheless, using this specification, the months between November 2000 and June 2003 are all classified in the two lowest quartiles of lagged market performance, neither of which have significant returns. In sum, our results indicate that the poor performance of momentum strategies during the sample period can be attributed to the negligible returns obtained during the 2001 to 2004 subperiod. This in turn can be attributed to the poor contemporaneous performance of the stock market as a whole. As such, we show that momentum returns can be adversely affected by stock market-wide phenomena.

3.2.3 Momentum and economic states

The results in the previous section support the argument that market states influence momentum returns. However, the observed relationship between momentum returns and market states could be simply accounting for the relationship with the wider

macroeconomic environment. Indeed, aggregate stock market movements are often regarded as a forward-looking indicator for movements in the economy. Chordia and Shivakumar (2002) examine this relationship, finding that momentum returns are related to the business cycle in the US. To address this concern, we use seasonally adjusted real GDP growth to sort our sample period into states of positive and negative GDP growth. If momentum is related to macroeconomic risk, we would expect to observe insignificant or negative momentum returns when that risk is realised i.e. in periods of negative GDP growth.

Panel B of Table 3 documents the return of a 6/1/6 strategy in expansionary and contractionary periods. We find that momentum returns are positive in both periods of economic expansion and contraction, with average returns of 1.09% and 2.39% respectively. The average return during positive GDP growth is significant. Although momentum returns are not significant during periods of negative GDP growth, this is not unexpected given that there are only nine months where GDP growth was negative.

Within our sample, the magnitude of momentum returns during periods of negative growth suggests that these results should be treated with caution. Indeed, the periods of economic contraction within the sample occur entirely during the tech boom. As such, the momentum returns observed during periods of negative GDP growth are entirely driven by the predominantly large momentum returns recorded during the tech boom. A better indication of the relationship between macroeconomic states and momentum returns may be gained by sorting the sample period into GDP growth quartiles. From Table 3, momentum returns are -0.15% , 3.37%, -0.88% and 2.23% for the lowest to highest quartiles of GDP growth. The correlation between GDP and

momentum returns is only 0.075. This suggests that momentum returns are not related to GDP growth, with if anything, momentum returns slightly higher in periods of below average GDP growth.

Finally, we note that classifying months according to ex-post realised GDP growth is not useful to investors when making investment decisions. Petkova and Zhang (2002) argue that unexpected poor economic news is more likely to arrive in the early stages of a recession. Conversely, poor economic news is more likely to be expected in the latter stages of a recession. As such, they argue that while a high risk stock is likely to experience poor performance during periods of unexpectedly high risk, this is not the case during periods of high expected risk. To address this concern, one could follow Griffin et al. (2003) and examine momentum returns in the first quarter of a recession. However, given that no consecutive periods of negative GDP growth were recorded over the sample period, the length of recessions within our sample precludes such an analysis.

Nevertheless, in contrast to Chordia and Shivakumar (2002), our results suggest that momentum returns in our sample are not related to the business cycle.²⁵ As such, the observed relationship between market states and momentum returns is not a result of macroeconomic risk. Finally, we stress that we are not contending that momentum is dependent on the market states but that *in our sample* the profitability of momentum strategies is related to the state of the market. The length of our sample period precludes

²⁵ As noted in the literature review, Griffin et al. (2003) and Cooper et al. (2004) also find no relationship between macroeconomic risk and momentum returns. Griffin et al. argue that the results of Chordia and Shivakumar (2002) are a result of methodological flaws, whilst Cooper et al. find that microstructure biases may explain Chordia and Shivakumar's results.

any general inference about the relationship between momentum and the state of the market or the economy.

3.3 Sub sample Analyses

Consistent with prior literature, the previous section has shown that momentum returns in the period 1993 to 2004 are linked to the performance of the stock market. Consistent with this relationship, momentum returns in the 2001 to 2004 subperiod are negligible. As the stock market crash of 2000 was most pronounced in the smaller, high-tech firms, it is of interest whether the poor momentum return in the 2001 to 2004 subperiod is confined to smaller or high-tech stocks.

3.3.1 Size subsamples

As a robustness check, we first examine the returns of momentum strategies that exclude all stocks with market capitalisations that would place them in either of the two smallest NYSE market capitalisation deciles. Given that momentum returns are not significant within our sample, eliminating small firms serves to examine whether small or illiquid firms or bid-ask bounce are contaminating our results.

Panel A of Table 4 presents the returns of the 6/1/6 strategy where the small capitalisation stocks are eliminated from the sample. The results reveal that eliminating small stocks from the sample leads to a significant momentum return of 1.56% for the 6/1/6 strategy. Not only is this return less volatile than that of the full sample, but it is also substantially larger. This suggests that momentum returns in our sample are weaker in small capitalisation stocks, in contrast to prior studies.

Insert Table 4 here

To examine the relationship between momentum and firm size in detail, we sort firms by size and construct momentum portfolios for each size subsample. Table 4 Panels B, C and D report the average returns of a 6/1/6 strategy for small, medium and large firms respectively. Consistent with Jegadeesh and Titman (1993), we find that momentum returns are somewhat related to firm size. As implied by Panel A, momentum returns are weaker for small stocks, with a marginally significant return of 1.15%. For the medium and large firms, momentum returns are significant and 1.70% and 1.46% respectively. These results indicate that the momentum effect has continued to exist in the period of 1993 to 2004 for all size sub samples. The widespread use of momentum strategies subsequent to the initial Jegadeesh and Titman (1993) finding of momentum has not eliminated momentum returns.

Consistent with prior literature, we find that momentum returns are greatest for medium-size firms. Nonetheless, in contrast to Jegadeesh and Titman (1993, 2001), we find that momentum returns are greater in large firms than in small firms. The momentum return for large firms within our sample is almost twice the 0.75% reported by Jegadeesh and Titman (1993). An analysis of the subperiod results shows that the large firm momentum strategy earns significant returns in only the 1997 to 2000 subperiod. Finally, we note that the return of the large firm momentum strategy is predominantly driven by the loser portfolio. For large firms, the winners outperform the value-weighted (equal-weighted) index by 0.37% (-0.209%) per month, whereas the

losers underperform the value-weighted (equal-weighted) index by 1.09% (1.66%) per month.

The subperiod returns show that while small and mid-size momentum strategies were profitable during the 1993 to 1996 and 1997 to 2000 subperiods, their returns over the 2001 to 2004 subperiod are not significantly different from zero. The insignificant momentum return for the 2001 to 2004 subperiod is robust to the choice of ranking and holding periods, with many of the small firm strategies earning negative returns over this subperiod. The poor momentum return over the 2001 to 2004 subperiod is not restricted to any size subsample. Nevertheless, the returns for small firms during this subperiod are uniformly lower than that of both medium and large firms.

3.3.2 Exchange subsamples

The inclusion of Nasdaq-listed stocks in the sample is rare in the momentum literature. To assess whether our results are due to the inclusion of Nasdaq-listed stocks in our sample, we examine the returns of a subsample including only stocks listed on the NYSE or AMEX, and one consisting entirely of stocks listed on Nasdaq.

Insert Table 5 here

Table 5 documents the return of the 6/1/6 strategy for the two exchange-based subsamples. Consistent with the results for the size sorts, we find that momentum returns are larger for the NYSE/AMEX subsample, with the NYSE/AMEX subsample generating more significant returns for all subperiods. Over the entire period, a momentum strategy

that selects stocks from the NYSE and AMEX returns a significant 1.18% per month, while a momentum strategy restricted to Nasdaq stocks returns an insignificant 1.08% per month. Pertinently, we find that within the NYSE/AMEX subsample momentum returns are significant for the entire sample period and highly significant for the 1993 to 1996 and 1997 to 2000 subperiods. In unreported results, the length of the holding period does not materially affect these results, with all but one strategy using a six-month ranking period significant over the sample period, and all highly significant over the 1993 to 2000 subperiod. The results for the NYSE/AMEX subsample are thus more consistent with prior literature than the full sample results, for which only 1 of the 32 momentum strategies is (marginally) significant. We also find that the NYSE/AMEX momentum strategy has a substantially lower standard deviation of returns (6.00%) relative to the full sample (8.43%).

3.4 ‘front-running’ Momentum Strategies

Table 6 presents the average monthly returns for the losers (R1), winners (R10) and the winner minus loser portfolio for strategies with and without a gap period of one month. Across the entire sample, we find that a ‘front-running’ momentum strategy generates zero-cost portfolio returns of approximately one percent a month. As with the month-end strategies, there is substantial variation in the profitability of individual ‘front-running’ momentum strategies.

In contrast to the month-end strategies, the returns of a number of the ‘front-running’ strategies are significant across the entire period. For example, the 6/1/3 strategy generates a significant positive return when a ‘front-running’ evaluation date is used, whereas the corresponding month-end strategy produces an insignificant return. In

general, the momentum returns of ‘front-running’ strategies are broadly similar to, if not slightly greater than their equivalent month-end momentum strategies. In addition, the ‘front-running’ strategies are less volatile than the month-end strategies, with a lower standard deviation for all of the 32 strategies examined.

The uniformly lower standard deviation of the ‘front-running’ strategies is particularly interesting, as it indicates that using a ‘front-running’ portfolio evaluation date results in less volatile returns. This suggests that there is increased return volatility at the end of the month, consistent with institutional trading impacting on returns.

Insert Table 6 here

When using a three-month ranking period, the returns of ‘front-running’ momentum strategies are higher than the corresponding month-end strategies for all investment horizons with or without a gap period. The excess return of the ‘front-running’ momentum strategies over their month-end counterparts ranges from an annualised return of 0.34% to 2.77%. The magnitude of these differences suggests that using a ‘front-running’ strategy may generate economically but statistically significant returns in excess of a month-end strategy with lower volatility. The *t*-statistic of all such strategies are higher when using a ‘front-running’ strategy, and of the strategies that employ a three-month ranking period, the returns of two are statistically significant using a ‘front-running’ strategy. In contrast, none of the corresponding month-end strategies generated significant returns. Indeed, consistent with our hypothesis, the ‘front-running’ effect appears to be strongest in momentum strategies that use a three-month ranking

period. Conversely, the returns of ‘front-running’ strategies that use 6, 9 or 12 month ranking criteria are virtually identical to the equivalent month-end strategy.

3.5 The Structure of ‘front-running’ Momentum Returns

It is possible that the insignificant differences of returns in Table 6 are caused by the returns of the ‘front-running’ portfolio being contaminated by stocks that are not affected by institutional momentum trading. If this is the case, restricting the sample to those stocks likely to experience momentum trading should lead to a more distinct ‘front-running’ effect. To this end, prior research has suggested that momentum trading is predominantly a winners phenomena (Badrinath and Wahal, 2002), which is itself primarily a large firm phenomena (Grinblatt et al., 1995). As such, one would expect that the ‘front-running’ effect would be greater amongst large cap winners.

We examine the returns of month-end and ‘front-running’ momentum strategies for a subsample of large stocks. As previously noted, prior studies suggest that the ‘front-running’ effect would be strongest when a three-month ranking period is used. As our results also indicate that the ‘front-running’ effect is strongest when a three-month ranking period is used, we limit our analysis in this and following sections to ‘front-running’ strategies that use a three-month ranking period.

The results in the leftmost of Tables 7 indicate that the ‘front-running’ effect is not more pronounced in large capitalisation stocks nor can it be attributed to large capitalisation winners. Rather, we find that ‘front-running’ strategies have lower returns, standard deviations and t -statistics amongst large capitalisation stocks as compared to analogous month-end strategies. Interestingly, the results also indicate that the

momentum effect is not present for large stocks within the period of 1993 to 2004 when using a three-month ranking criterion. This contrasts with the results for a 6/1/6 strategy presented in the prior section and implies that the large firm momentum effect is sensitive to the specification of the ranking period.

Our findings suggest that momentum trading does not affect the return series of large capitalisation stocks as predicted. However, we note that momentum trading may not exert significant price pressure on large capitalisation stocks. In addition, the age of the Grinblatt et al. (1995) findings raise the question of their applicability to the market of the recent past. Recently, momentum trading is more likely to impact on small capitalisation stocks, as these are more likely to be affected by institutional trading imbalances. Indeed, Wermers (1999) find that both herding and the relationship between institutional ownership and contemporaneous returns is strongest in small stocks. Consequently, we also investigate ‘front-running’ momentum returns in small and medium capitalisation stocks to examine whether ‘front-running’ leads to greater returns. These results are reported in the rightmost columns of Table 7.

Insert Table 7 here

For the small capitalisation stocks, we find that a ‘front-running’ strategy generates consistently higher returns, with the excess return peaking at 0.23% for the 3/0/3 strategy. Further, the returns of the ‘front-running’ strategies are more significant than the month-end strategies, with the 3/1/12 strategy (marginally) significant only when a ‘front-running’ strategy is used. Similarly, the 3/1/6 and 3/1/9 strategies are significant

when a ‘front-running’ strategy is used, whereas they are only marginally significant for the month-end strategies. In general, the difference in returns is insignificantly larger for strategies that form the portfolios immediately, with an average return difference of 0.13% per month compared to 0.07% when the portfolio formation is delayed by a month. An analysis of the return difference shows that the superior returns of the ‘front-running’ strategies are primarily driven by the lower return of the loser portfolio relative to the month-end momentum strategies. This is in contrast to our hypothesis.

For the medium size firms, ‘front-running’ momentum returns are similar to the month-end counterparts for the medium size firms. This suggests that a ‘front-running’ strategy only generate an excess return relative to a month-end strategy for small firms. As bid-ask bounce and infrequent trading is more prevalent for such stocks, we cannot rule out that the increased returns generated by a ‘front-running’ strategy are due to stock illiquidity. The ‘front-running’ strategies have lower standard deviations in all size subsamples, and for all strategies examined within each size subsample.

4. Conclusions

We find large variations in returns from momentum strategies. Momentum strategies did not earn significant returns during the period of 1993-2004 which was due to their particularly poor performance over the period from 2001-2004. A sample period specific finding is that returns of momentum portfolios restricted to NYSE and AMEX stocks are stronger than when Nasdaq stocks are included. We find that the previously documented large firm momentum effect is sensitive to the momentum strategy examined, and is in our sample driven by the abnormal returns of large Nasdaq stocks. Further we evaluate momentum strategies that do not adhere to the end of month

portfolio formation universally used in the academic literature. We form portfolios one week prior to the end of month and call them ‘front-running’ momentum portfolios. Consistent with institutional momentum trading affecting end of month returns and volatility, we find that ‘front-running’ a momentum strategy generates economically significant returns in excess of a month-end strategy for small firms. Pertinently, the returns of a ‘front-running’ strategy are consistently less volatile than that of an equivalent month-end strategy.

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Figure 1: Time Series of Momentum Returns

Figure 6.1 shows the time-series of one-month returns from a momentum strategy based on 6-month lagged returns, with no skip period and a holding period of 6 months (i.e. a 6/0/6 strategy).

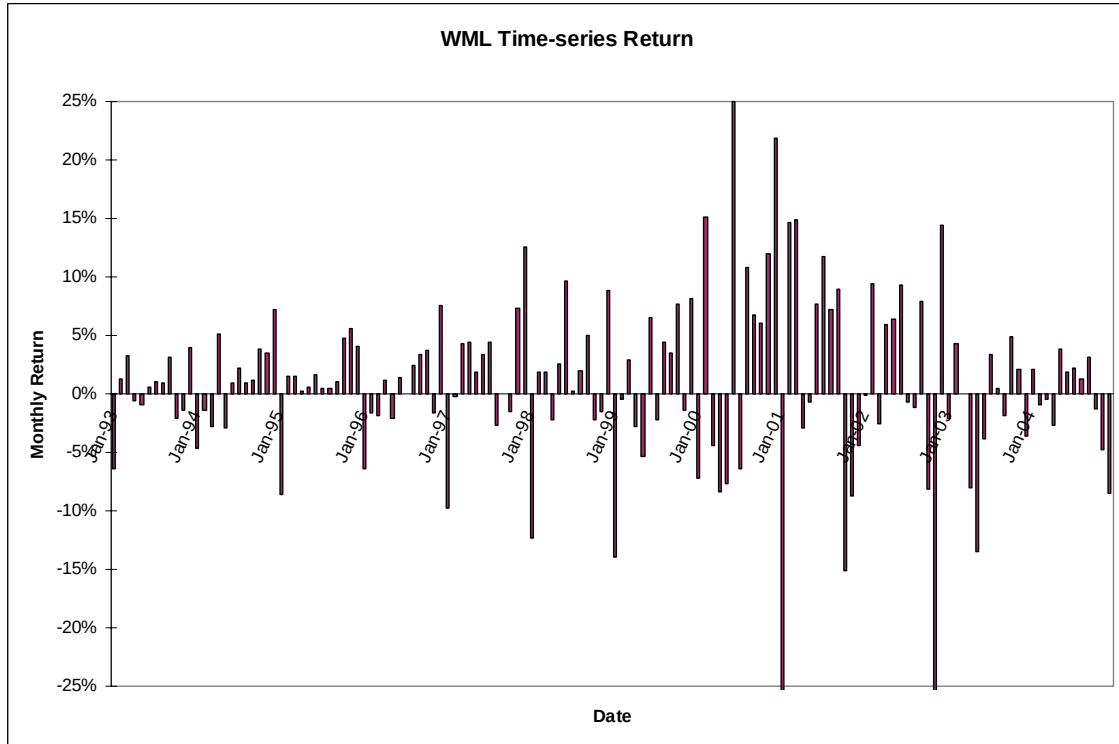


Table 1: Momentum returns with one month skip period and with immediate portfolio formation

Reported are the returns of the momentum portfolios based on J -month lagged returns and held for K months. The winner portfolio is an equally weighted portfolio of stocks in the highest lagged return decile. The loser portfolio is an equally weighted portfolio of stocks in the lowest return decile. The average monthly returns of these portfolios are presented along with the t-statistics in brackets. *, ** and *** denote significance at the 10, 5, and 1 % level.

Panel B: No skip period					
Ranking Period (J)	Portfolio	Holding Period (K)			
		3	6	9	12
3	Winner (R10)	1.378% (2.12)**	1.525% (2.47)**	1.438% (2.41)**	1.343% (2.23)
	Loser (R1)	1.097% (1.09)	0.825% (0.89)	0.829% (0.81)	0.848% (1.05)
	Winner - Loser	0.281% (0.36)	0.700% (1.11)	0.610% (1.07)	0.494% (1.13)
6	Winner (R10)	1.887% (2.83)**	1.758% (2.85)***	1.609% (2.61)**	1.393% (2.27)
	Loser (R1)	0.874% (0.82)	0.748% (0.75)	0.784% (0.86)	0.892% (1.07)
	Winner - Loser	1.013% (1.13)	1.010% (1.35)	0.824% (1.29)	0.501% (0.98)
Panel B: One month skip period					
3	Winner (R10)	1.433% (2.24)**	1.528% (2.48)**	1.440% (2.41)**	1.282% (2.14)**
	Loser (R1)	0.584% (0.62)	0.602% (0.68)	0.680% (0.81)	0.796% (1.04)
	Winner - Loser	0.849% (1.24)	0.925% (1.56)	0.760% (1.53)	0.487% (1.27)
6	Winner (R10)	1.888% (2.92)***	1.765% (2.85)***	1.511% (2.46)**	1.286% (2.11)**
	Loser (R1)	0.521% (0.51)	0.592% (0.63)	0.725% (0.85)	0.874% (1.11)
	Winner - Loser	1.368% (1.65)	1.172% (1.67)*	0.785% (1.42)	0.412% (0.91)

Table 2: Subperiod Returns of Momentum Portfolios

Reported are the returns for momentum portfolios based on 6-month lagged returns, with a skip period of 1 month and a holding period of 6 months (i.e. a 6/1/6 strategy). This table reports the average momentum return within 4-year subperiods. The Equal Weighted Index is the returns on the equal-weighted index of all stocks over that period. The winner portfolio is an equally weighted portfolio of stocks in the highest lagged return decile. The loser portfolio is an equally weighted portfolio of stocks in the lowest return decile. The average monthly returns of these portfolios are presented along with the t-statistics in brackets. *, ** and *** denote significance at the 10, 5, and 1 % level.

Portfolio	1993-2004	Subperiod		
		1993-1996	1997-2000	2001-2004
Winner (R10)	1.758% (2.85)***	1.706% (2.57)**	1.756% (1.18)	1.831% (2.02)**
Loser (R1)	0.748% (0.75)	0.671% (0.82)	-0.636% (-0.43)	1.756% (0.77)
Winner - Loser	1.010% (1.35)	1.035% (2.31)**	2.393% (2.21)**	0.075% (0.04)
Equal Weighted Index	1.697%	1.290%	0.684%	1.737%

Table 3: Momentum Returns and Market Conditions

Reported are the returns for momentum portfolios based on 6-month lagged returns, with a skip period of 1 month and a holding period of 6 months (i.e. a 6/1/6 strategy). This table reports the average momentum return within 4-year subperiods. Panel A reports momentum returns in different stock market states based on lagged 1-year S&P500 returns. On the left are results in up- and down-markets, and on the right are averages based on quartiles of lagged returns. Panel B reports momentum returns in different economic states based on quarterly real adjusted GDP growth. On the left are results in positive and negative market states, and on the right are averages based on quartiles of GDP growth. The winner portfolio is an equally weighted portfolio of stocks in the highest lagged return decile. The loser portfolio is an equally weighted portfolio of stocks in the lowest return decile. The average monthly returns of these portfolios are presented along with the t-statistics in brackets. *, ** and *** denote significance at the 10, 5, and 1 % level.

<i>Panel A: Aggregate Stock Market State</i>						
Portfolio	Positive/Negative		Quartile			
	$r_m < 0$	$r_m > 0$	Lowest	2	3	Highest
Winner (R10)	0.237%	2.298%	0.025%	1.330%	3.065%	2.666%
	(0.22)	(3.12) ***	(0.02)	(0.78)	(2.69) ***	(3.35) ***
Loser (R1)	-0.775%	1.069%	-1.057%	0.767%	1.205%	1.473%
	-(0.25)	(1.48)	-(0.34)	(0.55)	(1.10)	(1.17)
Winner - Loser	1.012%	1.229%	1.082%	0.563%	1.860%	1.193%
	(0.43)	(2.50) **	(0.45)	(0.55)	(3.26) ***	(1.33)
<i>Panel B: Economic Growth State</i>						
Portfolio	Positive/Negative		Quartile			
	GDP < 0	GDP > 0	Lowest	2	3	Highest
Winner (R10)	-2.109%	2.028%	2.797%	-0.423%	1.872%	2.849%
	(3.22) ***	-(0.75)	(2.31) **	-(0.45)	(1.53)	(1.93) *
Loser (R1)	-4.494%	0.941%	2.943%	-3.794%	2.754% *	0.615%
	(1.17)	-(0.48)	(1.07)	-(2.57) **	(1.78)	(0.52)
Winner - Loser	2.385%	1.087%	-0.146%	3.370%	-0.882%	2.234%
	(1.91) *	(0.30)	-(0.06)	(3.70) ***	-(1.06)	(2.35) **

Table 4: Size-sub sample Returns of Momentum Portfolios

Reported are the returns for momentum portfolios based on 6-month lagged returns, with a skip period of 1 month and a holding period of 6 months (i.e. a 6/1/6 strategy). This table reports the average momentum return within small, medium and large stocks. Panel A also reports the returns of momentum portfolio for a strategy that eliminates all stocks with market capitalisations that would place it in either of the 2 smallest NYSE market capitalization deciles. The winner portfolio is an equally weighted portfolio of stocks in the highest lagged return decile. The loser portfolio is an equally weighted portfolio of stocks in the lowest return decile. The average monthly returns of these portfolios are presented along with the t-statistics in brackets. *, ** and *** denote significance at the 10, 5, and 1 % level.

Panel A: Returns Eliminating Small Capitalization Stocks				
Portfolio	1993-2004	Subperiod		
		1993-1996	1997-2000	2001-2004
Winner (R10)	1.456%	1.490%	2.040%	0.841%
	(2.10)**	(1.99)*	(1.20)	(0.88)
Loser (R1)	-0.108%	0.560%	-0.949%	0.071%
	-(0.13)	(0.92)	-(0.74)	(0.03)
Winner - Loser	1.564%	0.930%	2.989%	0.770%
	(2.28)**	(2.21)**	(2.26)**	(0.51)
Panel B: Small Cap stocks				
Portfolio	1993-2004	Subperiod		
		1993-1996	1997-2000	2001-2004
Winner (R10)	1.900%	1.797%	1.467%	2.438%
	(3.23)***	(2.72)***	(1.07)	(2.64)**
Loser (R1)	0.747%	0.712%	-0.594%	2.141%
	(0.75)	(0.80)	-(0.37)	(0.91)
Winner - Loser	1.153%	1.085%	2.061%	0.297%
	(1.68)*	(2.31)**	(2.32)**	(0.17)
Panel C: Medium Cap stocks				
Portfolio	1993-2004	Subperiod		
		1993-1996	1997-2000	2001-2004
Winner (R10)	1.475%	1.482%	1.814%	1.129%
	(1.99)**	(1.84)*	(1.01)	(1.08)
Loser (R1)	-0.225%	0.289%	-1.337%	0.381%
	-(0.25)	(0.43)	-(0.94)	(0.17)
Winner - Loser	1.700%	1.193%	3.151%	0.748%
	(2.41)**	(2.61)**	(2.39)**	(0.48)
Panel D: Large Cap stocks				
Portfolio	1993-2004	Subperiod		
		1993-1996	1997-2000	2001-2004
Winner (R10)	1.487%	1.546%	2.338%	0.586%
	(2.29)**	(2.28)**	(1.44)	(0.70)
Loser (R1)	0.032%	0.940%	-0.348%	-0.490%
	(0.04)	(1.71)*	-(0.31)	-(0.26)
Winner - Loser	1.455%	0.606%	2.686%	1.075%
	(2.13)**	(1.34)	(1.83)*	(0.80)

Table 5: Exchange-sub sample Returns of Momentum Portfolios

Reported are the returns for momentum portfolios based on 6-month lagged returns, with a skip period of 1 month and a holding period of 6 months (i.e. a 6/1/6 strategy). Panel A reports the average momentum return within a sample constructed using only NYSE and AMEX-listed stocks. Panel B reports the results from a sample of Nasdaq-listed stocks. The winner portfolio is an equally weighted portfolio of stocks in the highest lagged return decile. The loser portfolio is an equally weighted portfolio of stocks in the lowest return decile. The average monthly returns of these portfolios are presented along with the t-statistics in brackets. *, ** and *** denote significance at the 10, 5, and 1 % level.

<i>Panel A: NYSE/AMEX Sub sample</i>				
Portfolio	1993-2004	Subperiod		
		1993-1996	1997-2000	2001-2004
Winner (R10)	1.749%	2.070%	1.461%	1.718%
	(4.18)***	(3.62)***	(1.70)*	(2.35)**
Loser (R1)	0.571%	0.713%	-0.702%	1.715%
	(0.86)	(1.16)	-(0.70)	(1.07)
Winner - Loser	1.179%	1.357%	2.163%	0.003%
	(2.36)**	(3.74)***	(3.01)***	(0.00)
<i>Panel B: Nasdaq Sub sample</i>				
Portfolio	1993-2004	Subperiod		
		1993-1996	1997-2000	2001-2004
Winner (R10)	1.723%	1.508%	1.820%	1.842%
	(2.47)**	(2.12)**	(1.07)	(1.80)*
Loser (R1)	0.643%	0.776%	-0.679%	1.848%
	(0.62)	(0.86)	-(0.41)	(0.75)
Winner - Loser	1.080%	0.732%	2.499%	-0.006%
	(1.45)	(1.41)	(2.21)**	(0.00)

Table 6: Returns of ‘front-running’ Momentum Portfolios

Reported are the returns for ‘front-running’ momentum portfolios based on J -month lagged returns, with a holding period of K months. Panel A reports the average momentum return when the portfolios are formed immediately after the ranking period. Panel B reports the results when a 1-month gap is present between the ranking and holding periods. The winner portfolio is an equally weighted portfolio of stocks in the highest lagged return decile. The loser portfolio is an equally weighted portfolio of stocks in the lowest return decile. The average monthly returns of these portfolios are presented along with the t-statistics in brackets. *, ** and *** denote significance at the 10, 5, and 1 % level.

<i>Panel A: No gap period</i>					
Ranking Period (J)	Portfolio	Holding Period (K)			
		3	6	9	12
3	Winner (R10)	1.502% (2.35)**	1.576% (2.47)**	1.509% (2.42)**	1.397% (2.27)**
	Loser (R1)	0.992% (1.00)	0.782% (0.86)	0.816% (0.83)	0.837% (1.04)
	Winner - Loser	0.510% (0.72)	0.794% (1.41)	0.694% (1.36)	0.560% (1.45)
6	Winner (R10)	1.888% (2.81)***	1.783% (2.79)***	1.644% (2.59)***	1.418% (2.24)**
	Loser (R1)	0.826% (0.79)	0.747% (0.77)	0.796% (0.89)	0.910% (1.11)
	Winner - Loser	1.062% (1.32)	1.036% (1.56)	0.849% (1.54)	0.509% (1.15)
<i>Panel B: Delayed portfolio formation</i>					
Ranking Period (J)	Portfolio	Holding Period (K)			
		3	6	9	12
3	Winner (R10)	1.511% (2.29)**	1.576% (2.47)**	1.497% (2.42)**	1.316% (2.13)**
	Loser (R1)	0.587% (0.64)	0.591% (0.68)	0.687% (0.83)	0.800% (1.05)
	Winner - Loser	0.924% (1.49)	0.985% (1.83)	0.810% (1.84)	0.515% (1.53)
6	Winner (R10)	1.871% (2.80)***	1.786% (2.79)***	1.535% (2.42)**	1.309% (2.08)**
	Loser (R1)	0.539% (0.54)	0.625% (0.68)	0.764% (0.91)	0.917% (1.18)
	Winner - Loser	1.332% (1.78)*	1.161% (1.88)*	0.771% (1.61)	0.392% (0.99)

Table 7: Returns of ‘front-running’ Momentum Portfolios For Size-based Subsamples

Reported are the returns for strategies based on 3-month lagged returns, with a holding period of K months. Panel A reports the average returns for strategies that use a month-end portfolio evaluation date for small, medium and large firm subsamples. Panel B reports the corresponding returns for ‘front-running’ strategies. Panel C reports the difference of return between the ‘front-running’ and month-end strategies. The winner portfolio is an equally weighted portfolio of stocks in the highest lagged return decile. The loser portfolio is an equally weighted portfolio of stocks in the lowest return decile. The average monthly returns of these portfolios are presented along with the t-statistics in brackets. *, ** and *** denote significance at the 10, 5, and 1 % level using t and paired t -tests.

Panel A: Month-end portfolios, no gap period												
	Small Cap				Mid Cap				Large Cap			
Holding period/ Portfolio	3	6	9	12	3	6	9	12	3	6	9	12
Winner (R10)	1.186% (1.86)*	1.715% (2.84)***	1.474% (2.73)***	1.438% (2.40)**	1.829% (2.45)**	1.210% (1.68)*	1.370% (1.65)	1.170% (1.71)*	1.568% (2.29)**	1.080% (1.69)*	1.320% (1.84)*	1.210% (2.01)**
Loser (R1)	1.058% (0.96)	0.911% (0.91)	0.909% (0.92)	0.942% (1.11)	1.184% (1.27)	0.516% (0.59)	0.395% (0.01)	0.373% (0.45)	1.147% (1.51)	0.644% (0.90)	0.599% (0.46)	0.619% (0.92)
Winner - Loser	0.128% (0.16)	0.804% (1.25)	0.565% (1.00)	0.496% (1.15)	0.645% (0.89)	0.693% (1.18)	0.976% (1.71)*	0.797% (1.75)*	0.421% (0.57)	0.437% (0.79)	0.721% (1.36)	0.591% (1.32)

Panel B: ‘front-running’ portfolios, no gap period												
	Small Cap				Mid Cap				Large Cap			
Holding period/ Portfolio	3	6	9	12	3	6	9	12	3	6	9	12
Winner (R10)	1.319% (2.04)**	1.769% (2.78)***	1.561% (2.50)**	1.494% (2.37)**	1.810% (2.52)**	1.182% (1.67)*	1.355% (2.01)**	1.179% (1.76)*	1.517% (2.47)**	1.072% (1.70)*	1.291% (2.18)**	1.164% (2.00)**
Loser (R1)	0.965% (0.87)	0.853% (0.86)	0.898% (0.96)	0.930% (1.10)	1.124% (1.26)	0.557% (0.66)	0.413% (0.49)	0.379% (0.47)	1.266% (1.74)	0.779% (1.13)	0.677% (0.98)	0.658% (1.00)
Winner - Loser	0.354% (0.47)	0.917% (1.56)	0.663% (1.27)	0.564% (1.45)	0.687% (1.11)	0.625% (1.23)	0.942% (1.92)*	0.800% (2.08)**	0.250% (0.41)	0.293% (0.58)	0.614% (1.29)	0.506% (1.24)

Table 7 (cont): Returns of ‘front-running’ Momentum Portfolios For Size-based Subsamples

<i>Panel C: Month-end portfolios one month gap period</i>												
	Small Cap				Mid Cap				Large Cap			
Holding period/ Portfolio	3	6	9	12	3	6	9	12	3	6	9	12
Winner (R10)	1.636%	1.718%	1.618%	1.491%	1.190%	1.214%	1.127%	0.950%	0.910%	1.081%	1.111%	0.976%
	(2.60)**	(2.85)***	(2.73)***	(2.49)	(1.58)	(1.69)*	(1.65)	(1.40)	(1.37)	(1.69)*	(1.84)*	(1.63)
Loser (R1)	0.721%	0.733%	0.805%	0.902%	-0.009%	-0.045%	0.007%	0.157%	0.263%	0.250%	0.320%	0.481%
	(0.70)	(0.77)	(0.92)	(1.13)	-(0.01)	-(0.05)	(0.01)	(0.20)	(0.37)	(0.36)	(0.46)	(0.73)
Winner - Loser	0.915%	0.984%	0.813%	0.589%	1.199%	1.259%	1.120%	0.793%	0.647%	0.831%	0.791%	0.495%
	(1.33)	(1.66)*	(1.68)*	(1.59)	(1.86)*	(2.13)**	(2.12)**	(1.93)*	(0.99)	(1.46)	(1.59)	(1.18)

<i>Panel D: ‘front-running’ portfolios one month gap period</i>												
	Small Cap				Mid Cap				Large Cap			
Holding period/ Portfolio	3	6	9	12	3	6	9	12	3	6	9	12
Winner (R10)	1.699%	1.770%	1.679%	1.512%	1.191%	1.183%	1.122%	0.941%	0.952%	1.067%	1.099%	0.960%
	(2.57)**	(2.78)***	(2.67)***	(2.40)**	(1.61)	(1.68)*	(1.66)*	(1.40)	(1.47)	(1.70)*	(1.85)*	(1.63)
Loser (R1)	0.657%	0.707%	0.802%	0.897%	0.135%	0.051%	0.064%	0.218%	0.490%	0.389%	0.386%	0.530%
	(0.65)	(0.75)	(0.92)	(1.12)	(0.16)	(0.06)	(0.08)	(0.28)	(0.72)	(0.57)	(0.57)	(0.82)
Winner - Loser	1.042%	1.063%	0.877%	0.615%	1.056%	1.132%	1.058%	0.724%	0.461%	0.677%	0.713%	0.430%
	(1.64)	(1.98)**	(2.00)**	(1.88)*	(1.79)*	(2.16)**	(2.35)**	(2.03)**	(0.80)	(1.29)	(1.55)	(1.09)